CS416 Project 3: Implement a Routing Protocol for the Virtual Network

Learning objectives:

1. Explain how link state and distance vector routing protocols work.
2. Implement a dynamic routing protocol in Java.

What to submit:

1. All java source files.
2. The config file.
3. A brief note summarizing:
   a. Each member's attendance at group meetings, including in-class collaboration sessions and any additional meetings scheduled by the group.
   b. Each member's contribution to the project as agreed upon by all members.

(Each group only needs to submit one copy on Canvas. Please include a submission comment listing all group members.)

Project Instructions:

1. In this project, you will extend the **Router** class you developed in Project 2 to enable dynamic routing.
   a. Remove all hardcoded forwarding tables from the previous project. Routers' forwarding tables must now be automatically generated by the routing protocol.

2. You may choose to implement either link state routing or distance vector routing. Only one is required.

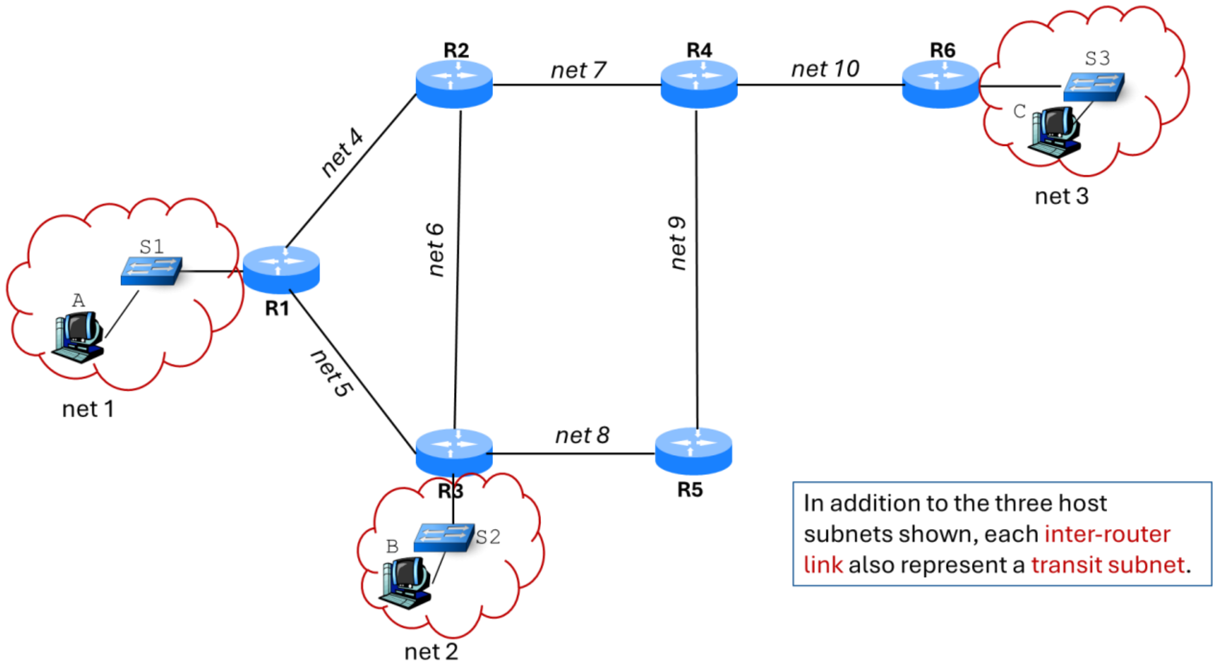
3. To simplify the implementation, we make the following assumptions:
   - Each link has a uniform cost of 1.
   - No link or router failures – your implementation does not need to handle these scenarios.
   - The network is perfectly reliable (i.e., acknowledgments and retransmissions are not required for link state routing)

4. Note that there are now two types of packets in this virtual network:
   a. **User packets**: generated by hosts and containing user messages
   b. **Routing updates**: generated by routers containing LSAs (link state routing) or DVs (distance vector routing)

   To differentiate these types, we can add a binary flag to each packet.
   This flag should be set by the sender (either a host or a router) when generating the packet.

Testing:

Use this network topology to test your implementation:



Ensure your config file includes all 10 subnets in this network:

- 3 host subnets
- 7 transit subnets between routers

Grading:

This project is worth a total of 100 points, distributed as follows:

- 70 points for implementation (detailed rubric will be included in the demo score sheet)
- 30 points for participation & contribution:
  - Attendance at all collaboration sessions, both in class and as scheduled by your group (10 points)
  - Attendance at the demo session. (10 points)
  - Meaningful individual contributions to the project. (10 points)

Demo session: you will be required to give a live demonstration of your work, which will be the primary basis for the implementation grade. A demo score sheet will be provided separately.

Partial credit: If your implementation is only partially correct, you will receive partial credit proportional to the level of functionality achieved.